



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to RJNS TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

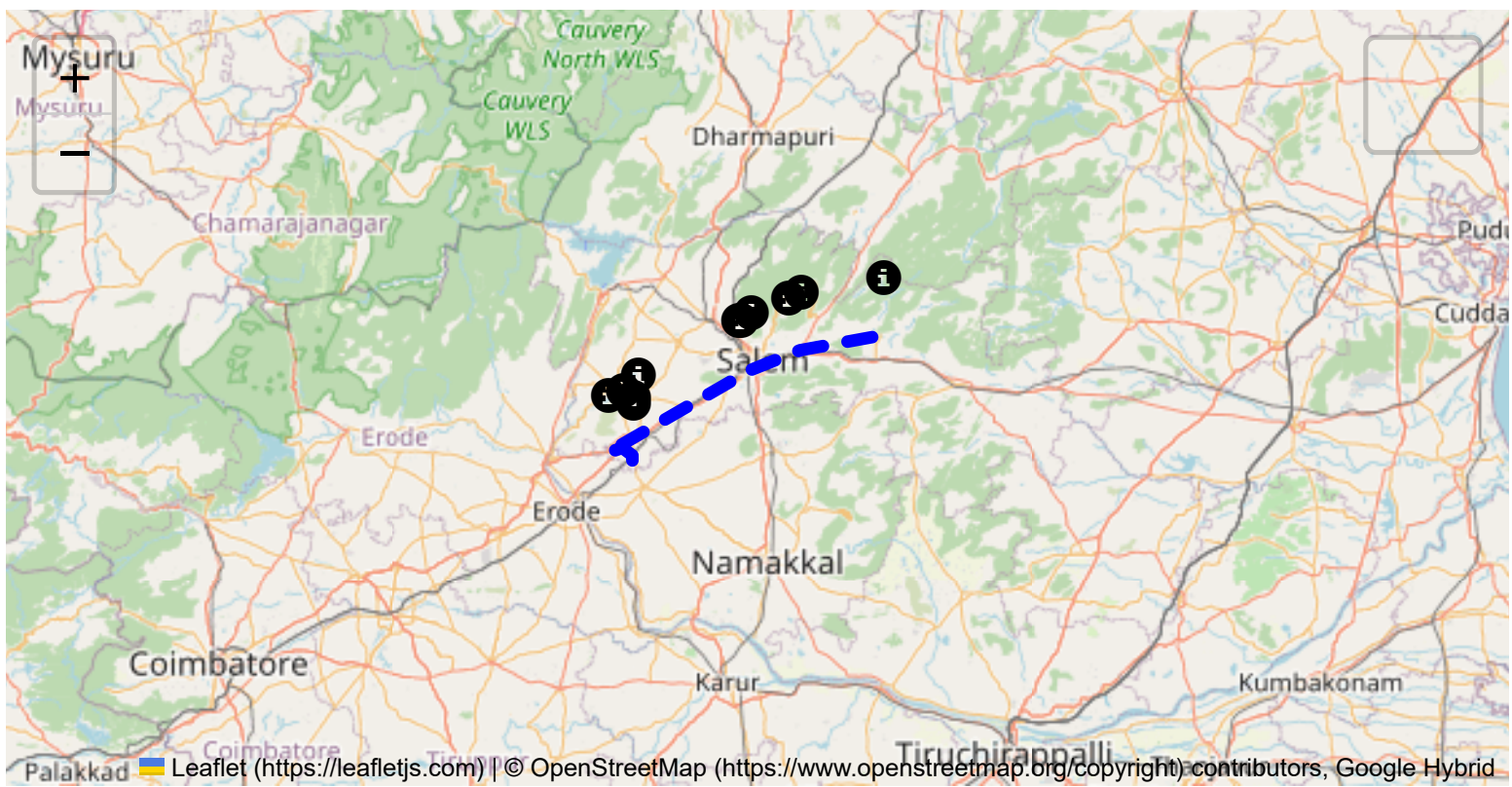
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

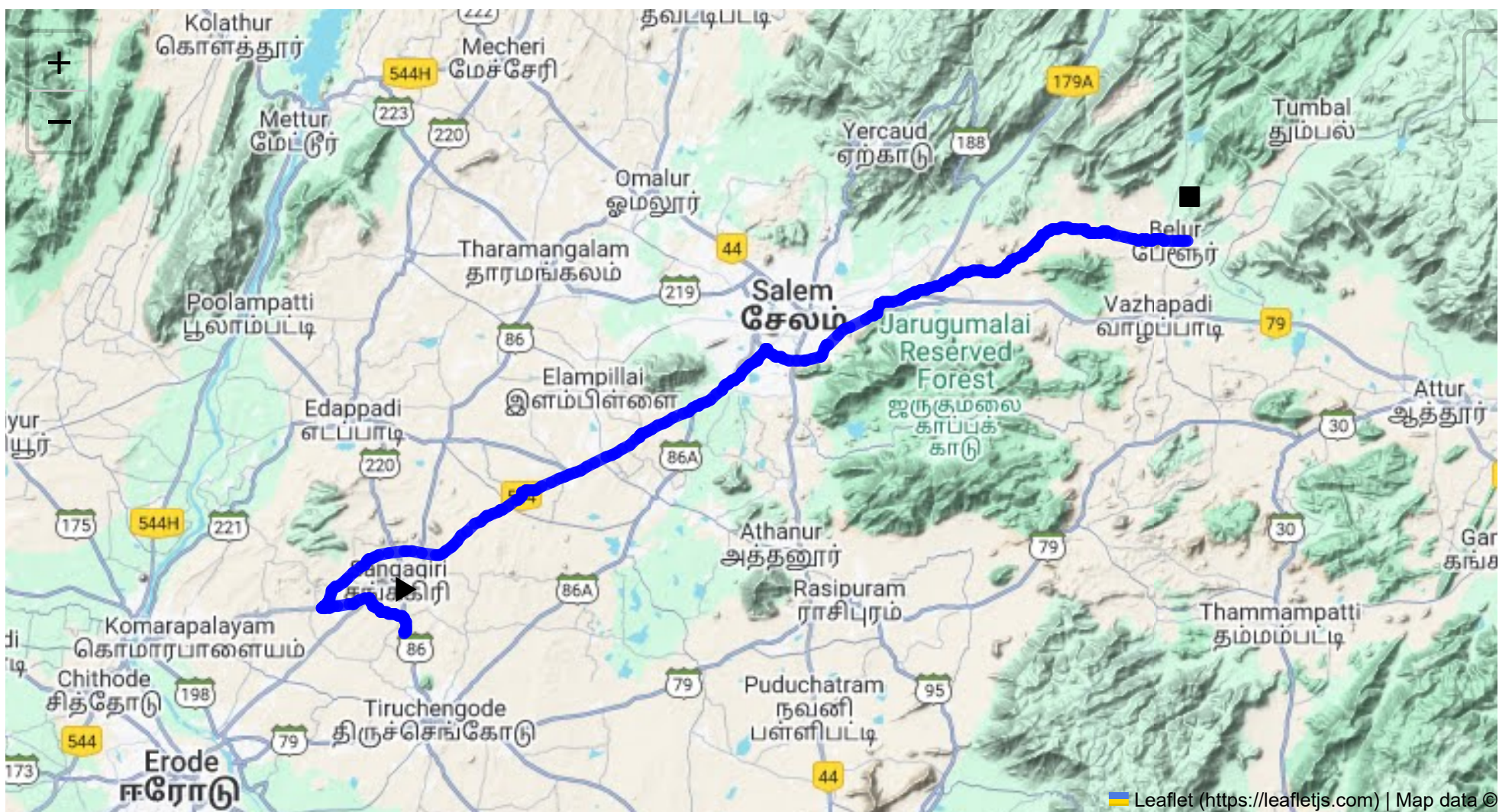
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 85.80 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours
Start: (11.4381, 77.8734)
End: (11.7072, 78.421811)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to PC4C+PMH, Belur, through the specified points, primarily follows regional roads in Tamil Nadu, India. It connects smaller towns and rural areas, passing through significant urban centers like Salem. Key highways include NH544 and key regional roads like Bhavani Main Road. The route covers diverse terrains, including urban streets, highways, and rural passages.

2. Typical Weather Conditions and Potential Weather-related Hazards

Tamil Nadu experiences a tropical climate with hot summers (March to June), significant monsoon rains (June to September), and mild winters (November to February). During monsoon season, heavy rains may cause waterlogging in low-lying areas, increasing the risk of hydroplaning and deteriorating road conditions. In summer, high temperatures may affect vehicle engines and increase tire blowout risks.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic congestion is common during peak hours (8-10 AM and 5-7 PM), particularly around urban centers like Salem.
- **Congestion-prone Areas:** Key areas of congestion include the intersections near Salem, where local and through traffic mix, and smaller towns where roads narrow or converge.

4. Assessment of Road Quality and Infrastructure

- **Highways (e.g., NH544):** Generally well-maintained with clear signage, though occasional potholes can be found.
- **Regional Roads (e.g., Bhavani Main Road):** Mixed quality, with some sections prone to wear that can worsen during monsoons. Road shoulders may be narrow or absent in rural areas.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Routes:**
 - Use the Salem-Kochi Highway for quicker detours.
 - Local road networks in Salem provide additional routing options but be wary of smaller roads unsuitable for heavy vehicles.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is governed by Indian regulations, requiring special permits and adherence to designated routes and times (often restricted during peak traffic hours). Ensure compliance with the Motor Vehicles Act and observe local posting requirements.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Incidents involving heavy vehicles on these roads often include overturns and collisions, particularly near congested urban centers and during adverse weather. Historical data indicates higher risks in Salem due to traffic density.

8. Environmental Considerations and Sensitive Areas

The route passes through agricultural areas that may have restrictions on pollution and noise. Urban and residential zones require careful consideration of local environmental impact laws.

9. Analysis of Communication Coverage

Most of the route, particularly near urban areas, enjoys good mobile network coverage. However, rural stretches may experience occasional dead zones, though these are becoming less frequent with network expansions.

10. Estimated Emergency Response Times for Different Route Segments

- **Urban Areas (e.g., Salem):** Response times can range from 15-30 minutes due to proximity to services.
- **Rural Areas:** Response times can extend to 45 minutes or more, depending on road conditions and emergency service availability.

11. Overall Summary of Risk Assessment

The route presents moderate risk levels, primarily driven by weather conditions, road quality variances, and traffic congestion in urban areas. Effective risk mitigation strategies include choosing travel times outside peak hours, regular vehicle maintenance, compliance with safety regulations, and situational awareness for quick response to changing conditions. Emergency protocols should be reviewed occasionally, with driver awareness training focused on major points and potential hazards along the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45847, 77.85140	30 KM/Hr
7	Turn	Medium	11.45898, 77.85135	30 KM/Hr
8	Turn	Medium	11.46303, 77.84945	30 KM/Hr
9	Turn	Medium	11.46314, 77.84928	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45634, 77.81673	15 KM/Hr
12	Turn	High	11.61233, 78.10397	15 KM/Hr
13	Turn	High	11.61216, 78.10423	15 KM/Hr
14	Turn	High	11.63300, 78.12746	15 KM/Hr
15	Turn	Medium	11.65627, 78.20215	30 KM/Hr
16	Turn	Medium	11.65664, 78.20240	30 KM/Hr
17	Turn	Medium	11.65825, 78.20469	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Turn	Medium	11.66366, 78.20649	30 KM/Hr
19	Turn	Medium	11.67128, 78.22971	30 KM/Hr
20	Turn	High	11.67111, 78.23134	15 KM/Hr
21	Turn	Medium	11.67195, 78.23146	30 KM/Hr
22	Turn	Medium	11.68490, 78.26190	30 KM/Hr
23	Turn	Medium	11.68562, 78.26787	30 KM/Hr
24	Turn	Medium	11.68604, 78.26834	30 KM/Hr
25	Turn	Medium	11.68753, 78.27612	30 KM/Hr
26	Turn	Medium	11.70546, 78.31554	30 KM/Hr
27	Turn	High	11.71514, 78.35005	15 KM/Hr
28	Turn	Medium	11.71422, 78.35062	30 KM/Hr
29	Turn	High	11.71268, 78.35418	15 KM/Hr
30	Turn	High	11.71289, 78.35430	15 KM/Hr
31	Turn	High	11.71259, 78.35559	15 KM/Hr
32	Turn	Medium	11.71296, 78.35598	30 KM/Hr
33	Turn	Medium	11.71232, 78.36098	30 KM/Hr
34	Turn	Medium	11.71401, 78.36303	30 KM/Hr
35	Turn	High	11.70761, 78.41639	15 KM/Hr
36	Turn	High	11.70674, 78.41686	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
17	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
22	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
23	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
25	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
30	hospital	Government hospital,Belur	11.70532, 78.419538	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
28	school	Govt Hr Sec School	11.6754733, 78.2459724	30 km/h	Medium
29	school	St Mary's Hr Sec School	11.6872096, 78.2796818	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673

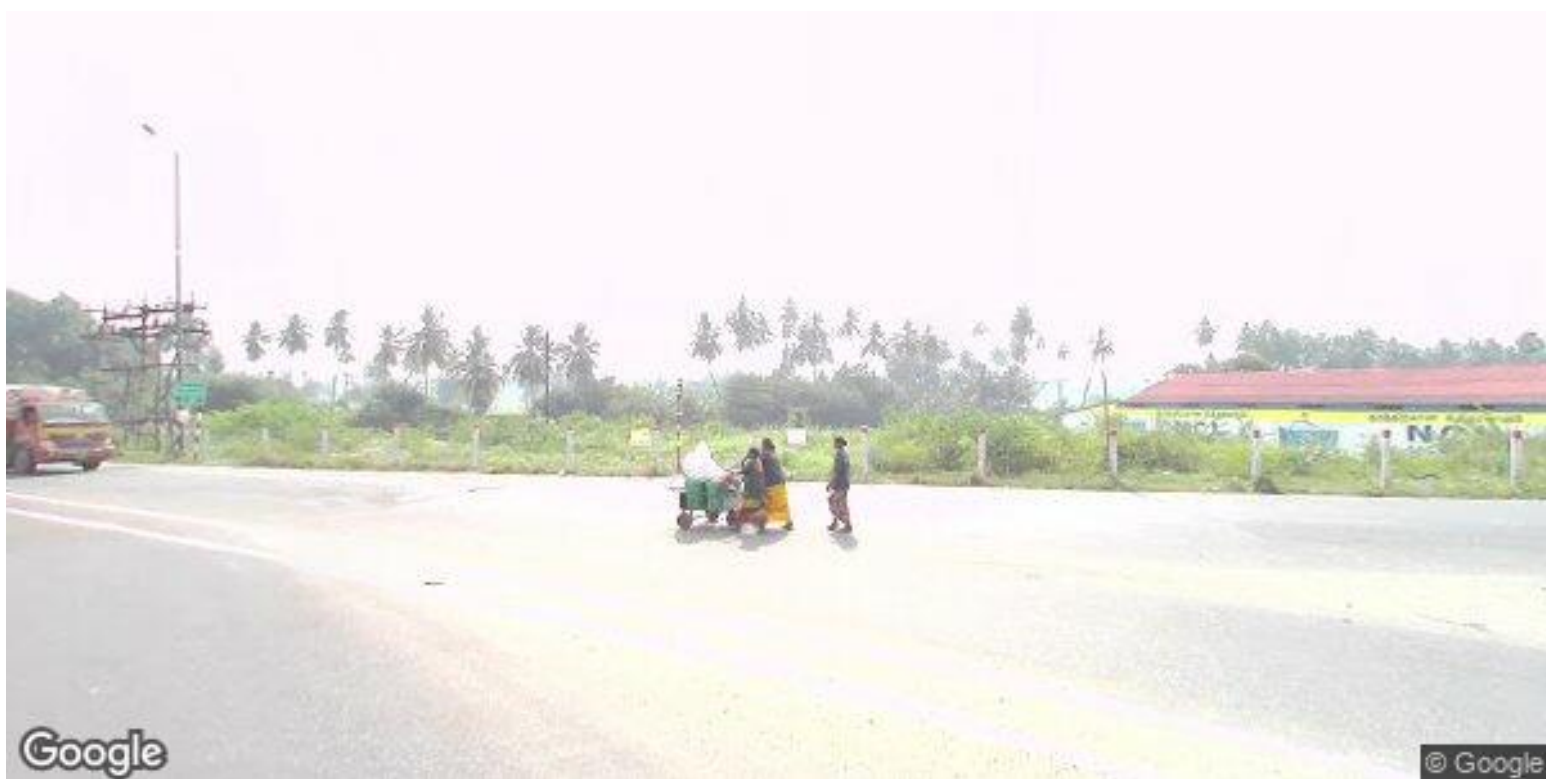


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61233, 78.10397



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61216, 78.10423



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63300, 78.12746



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65627, 78.20215



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65664, 78.20240



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65825, 78.20469



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66366, 78.20649



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67128, 78.22971



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67111, 78.23134



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67195, 78.23146



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68490, 78.26190



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68562, 78.26787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68604, 78.26834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68753, 78.27612



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70546, 78.31554



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70761, 78.41639



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70674, 78.41686

